

Orthogonal-Quotient Superloop Compression

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Abstract

Cloud-extracted traces were represented in RH-242 as a graded difference of physical loops and finite atomic counterloops. We give an exact cancellation-preserving grouping. The selected generalized root space is invariant, so its norm-one orthogonal quotient compression has power traces equal to the cloud-extracted traces. This avoids estimating the badly conditioned Riesz projection. A finite ordered-Schur audit verifies the identity on 17 tractable archived endpoints and shows that one-step quotient contraction still fails, motivating a block-power criterion.

1 Invariant root space and orthogonal quotient

Let $A \in \mathcal{S}_2(H)$ and let $\mathcal{S}$ be a finite algebraic multiset of isolated nonzero eigenvalues of A . Let E be the sum of their full generalized root spaces. Then E is finite dimensional and A -invariant. Write Π for the orthogonal projection onto E , $Q = I - \Pi$, and

$$C = QAQ|_{E^\perp} : E^\perp \longrightarrow E^\perp. \quad (1)$$

The complement $E^\perp$ need not be A -invariant: C is the orthogonal realization of the quotient operator, not a reducing restriction.

Theorem 1 (orthogonal quotient trace identity). *For every integer $n \geq 2$,*

$$\mathrm{Tr} A^n - \sum_{s \in \mathcal{S}} s^n = \mathrm{Tr} C^n. \quad (2)$$

The projection used in (1) satisfies $\|\Pi\| = 1$ whenever $E \neq \{0\}$; no norm bound for the oblique Riesz projection is required.

Proof. Relative to the orthogonal decomposition $H = E \oplus E^\perp$, invariance of E gives

$$A = \begin{pmatrix} A_E & X \\ 0 & C \end{pmatrix}.$$

Every power has the same block upper-triangular form and diagonal blocks A_E^n and C^n . Because $A \in \mathcal{S}_2$, these powers are trace class for $n \geq 2$. Hence $\mathrm{Tr} A^n = \mathrm{tr} A_E^n + \mathrm{Tr} C^n$. The finite-dimensional trace of A_E^n is the algebraic power sum $\sum_{s \in \mathcal{S}} s^n$, including multiplicity, which proves (2). The orthogonal projection norm is elementary. $\square$

This is precisely the RH-242 superloop cancellation [4] performed before taking absolute values. In an ordered Schur basis, closed block walks cannot contain the one-way coupling X : a walk that leaves the quotient block cannot return. The selected-block loops cancel the atomic counterloops, leaving only quotient loops.

Proposition 2 (quotient-kernel loops). *If $H = L^2(X, \mu)$ and A has a Hilbert–Schmidt kernel, then $B = QAQ$ has a Hilbert–Schmidt kernel b . For every $n \geq 2$,*

$$\mathrm{Tr} C^n = \mathrm{Tr} B^n = \int_{X^n} b(x_0, x_1) b(x_1, x_2) \cdots b(x_{n-1}, x_0) d\mu^n.$$

Thus the grouped representation remains a genuine signed or complex periodic-loop formula.

Proof. The Hilbert–Schmidt ideal is stable under bounded left and right multiplication, so B is Hilbert–Schmidt and vanishes on E in the orthogonal block form. The standard trace formula for products of Hilbert–Schmidt kernels gives the cyclic integral; its nonzero block is C^n . $\square$

2 Why this differs from a Riesz estimate

RH-232 found enormous Euclidean Riesz projection norms in the moving cloud [2]. Orthogonal quotient compression removes that norm from the fixed-operator trace identity, although it does not prove that the spaces E_σ vary uniformly.

Example 3 (unbounded obliqueness, fixed quotient). *For distinct λ, μ let*

$$A_M = \begin{pmatrix} \lambda & M \\ 0 & \mu \end{pmatrix}, \quad E = \mathrm{span}\{e_1\}.$$

The Riesz projection onto E is

$$P_\lambda = \begin{pmatrix} 1 & M/(\lambda - \mu) \\ 0 & 0 \end{pmatrix}, \quad \|P_\lambda\| = \sqrt{1 + |M/(\lambda - \mu)|^2},$$

which diverges with $|M|$. Yet $\Pi = \mathrm{diag}(1, 0)$, $C = [\mu]$, and

$$\mathrm{tr} A_M^n - \lambda^n = \mu^n$$

for every n , independently of M . Orthogonal grouping deletes purely cross-block nonnormal mass, but says nothing about nonnormality internal to C .

3 Finite ordered-Schur audit

We reconstruct every RH-222 endpoint of dimension at most 512 [1], Hardy-scale the matrix, and order its complex Schur form at the archived radial gap. The leading block contains Perron, parity, and the selected cloud; the trailing block is C . We compare its power traces with the projection-free RH-236 archive [3].

quantity	value
audited endpoints	17
selected block dimensions	6–25
rank mismatches	0
maximum Schur trace-partition error	3.99×10^{-15}
maximum RH-236 residual difference	6.31×10^{-12}
quotient Frobenius square	5.07–60.09
old-budget improvement factor	1.20–1.62
quotient one-step norm	1.277–2.346
first contractive power depth	3–7
maximum $\ C^{12}\ $	1.370×10^{-5}

The exact theorem explains why cross-block mass is irrelevant, but the finite numbers also prevent an overclaim: every audited one-step quotient norm exceeds one, and the Frobenius budget improves only modestly. Deep power contraction is striking finite evidence, not a uniform theorem.

4 Boundary and next condition

Theorem 1 is exact at each fixed noise once the algebraic selected subspace exists. It does not provide a noise-uniform method to identify that subspace, a uniform kernel bound, or a uniform contractive power. Only 17 of 32 archived endpoints were densely decomposed, and no interval certification was attempted. The next obligation is therefore an explicit m -block trace-envelope theorem and a uniform certificate for its hypotheses. The anchored selector obstruction of RH-244 remains separate.

Gate A remains open and Gates B–E are untouched. No Hilbert–Pólya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

- [1] Bin Wang. Rank-growing conjugate cloud atlas. 2026. RH-222 preprint.
- [2] Bin Wang. Biorthogonal riesz cloud projection. 2026. RH-232 preprint.
- [3] Bin Wang. Cloud-extracted trace moment atlas. 2026. RH-236 preprint.
- [4] Bin Wang. Cloud-extracted periodic superloop representation. 2026. RH-242 preprint.